Network Intrusion Detection System Powered by Machine Learning

Phase 2: Data Understanding

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1 Introduction

This report presents the Data Understanding phase of the CRISP-DM methodology applied to the **CIC-IDS2017** dataset. The objective of this phase is to gain a clear understanding of the dataset, including its structure and characteristics, and to identify any potential issues that will be addressed in the Data Preparation phase.

2 Dataset Overview

The **CIC-IDS2017 dataset** consists of network traffic data collected over five working days (Monday to Friday). The raw traffic was captured and then transformed into tabular data containing flow-level statistics (e.g., duration, packet counts, and rates) using CICFlowMeter. Comprising over 2.8 million rows and 79 columns, the dataset includes both benign and malicious traffic across 14 different attack types, such as DoS, Brute Force, DDoS, and SQL injection.

3 Methodology

The CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology is followed for the Data Understanding phase. CRISP-DM is a widely used framework for structuring machine learning projects. It consists of six phases:

- Business Understanding
- Data Understanding
- Data Preparation
- Modeling
- Evaluation
- Deployment

The Data Understanding phase involves loading the data, describing its structure and content, exploring its characteristics, and finally verifying its quality.

4 Data Loading

The dataset is distributed across eight CSV files stored in a shared data directory. Each file is assigned a short alias for use throughout the analysis, as presented in Table 1.

Table 1: Original filenames and their assigned aliases.

Original filename	**Alias**
`Monday-WorkingHours.pcap_ISCX.csv`	Monday
`Tuesday-WorkingHours.pcap_ISCX.csv`	Tuesday
`Wednesday-workingHours.pcap_ISCX.csv`	Wednesday
`Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv`	Thursday-AM
`Thursday-WorkingHours-Afternoon-Infilteration.pcap_ISCX.csv`	Thursday-PM
`Friday-WorkingHours-Morning.pcap_ISCX.csv`	Friday-AM
`Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv`	Friday-PM-DDoS
`Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv`	Friday-PM-PortScan

As shown in Table 1, each file is named after the day and session it corresponds to. The files are loaded into memory using pandas with the `low_memory=False` option to ensure consistent type inference across the full column range. All eight files are ingested successfully into a dictionary of DataFrames indexed by alias, which serves as the primary data structure for all subsequent analyses.

5 Data Description

This section examines the surface properties of the CIC-IDS2017 dataset in the order they are addressed in the analysis: file dimensions and column consistency, preliminary formatting corrections, label inventory, data combination, data types, duplicate columns, and finally the semantic interpretation of the features.

5.1 File Shapes and Column Consistency

The two following subsections verify that the eight files share a common structure, which is a prerequisite for safe concatenation and for using a single representative file in subsequent analyses.

5.1.1 File Shapes

The dataset comprises eight CSV files generated by CICFlowMeter from five days of network traffic captures. Table 2 presents the dimensions of each file after loading.

Table 2: Dimensions of each file after loading.

File alias	**Rows**	**Columns**
Monday	529 918	79
Tuesday	445 909	79
Wednesday	692 703	79
Thursday-AM	170 366	79
Thursday-PM	288 602	79
Friday-AM	191 033	79
Friday-PM-DDoS	225 745	79
Friday-PM-PortScan	286 467	79
Total	**2 830 743**	79

5.1.2 Column Consistency Check

As shown in Table 2, each file contains exactly 79 columns. A programmatic check confirms that all eight files share the same 79 column names in the same order, using the Monday file as the reference. This uniformity is an expected consequence of a single CICFlowMeter pipeline being applied to all captures and guarantees that a single file can serve as a representative sample for the schema-level analyses that follow, and that vertical concatenation into a unified DataFrame can be performed without column alignment issues.

5.2 Preliminary Formatting Corrections

Before any analysis is performed, two formatting defects are identified and corrected. Both corrections are purely cosmetic and do not alter any numerical feature value.

5.2.1 Whitespace in Column Names

Inspection of the raw CSV headers reveals that 65 out of 79 column names contain leading or trailing whitespace characters (for example, `' Destination Port'` in place of `'Destination Port'`). This is a documented artefact of CICFlowMeter's CSV export format. Left uncorrected, such whitespace causes `KeyError` exceptions when columns are accessed by name in pandas (e.g., `df['Label']` would fail). All column names are stripped across all eight files before any further step is performed.

5.2.2 Label Encoding Corruption

Three label values in the Thursday-AM file are found to contain mojibake characters: the hyphen separating "Web Attack" from the attack sub-type has been corrupted into the Unicode replacement character `U+FFFD`. Table 3 presents the affected labels alongside their corrected counterparts.

Table 3: Label encoding corrections applied to the Thursday-AM file.

Corrupted label	Corrected label
`Web Attack ? Brute Force`	`Web Attack - Brute Force`
`Web Attack ? XSS`	`Web Attack - XSS`
`Web Attack ? Sql Injection`	`Web Attack - Sql Injection`

As shown in Table 3, the three affected labels are exclusively Web Attack variants confined to the Thursday-AM file. The correction is applied via a direct string replacement before any label-based grouping or filtering is performed.

5.3 Label Inventory

A label inventory is compiled per file in order to verify that pandas has not introduced unexpected types and to document which attack classes are present in each capture session. The result confirms that the `Label` column is of type `object` across all files and that it is the only non-numeric column in the dataset. Table 4 presents the classes present in each file.

Table 4: Label inventory per file.

File alias	**Classes present**
Monday	BENIGN
Tuesday	BENIGN, FTP-Patator, SSH-Patator
Wednesday	BENIGN, DoS GoldenEye, DoS Hulk, DoS Slowhttptest, DoS slowloris, Heartbleed
Thursday-AM	BENIGN, Web Attack - Brute Force, Web Attack - XSS, Web Attack - Sql Injection
Thursday-PM	BENIGN, Infiltration
Friday-AM	BENIGN, Bot
Friday-PM-DDoS	BENIGN, DDoS
Friday-PM-PortScan	BENIGN, PortScan

As illustrated in Table 4, Monday contains exclusively benign traffic and serves as the baseline capture session. Each attack type appears in exactly one file.

5.4 Data Combination and Preview

Given the verified column consistency established in Section 5.1.2, the eight files are concatenated vertically into a single unified DataFrame `df_all` of **2,830,743 rows and 79 columns**. All subsequent analyses in this report operate on this combined dataset unless stated otherwise.

5.5 Data Types

The type composition of the dataset is assessed using the Monday file as a representative sample. Table 5 presents the distribution of column types across the 79 columns.

Table 5: Data type distribution across the 79 columns.

Type	**Count**	**Description**
`int64`	54	Integer-valued flow counters and TCP flag counts.
`float64`	24	Continuous flow statistics: rates, means, standard deviations.
`object`	1	The `Label` column, the sole non-numeric field.
Total	**79**	

As shown in Table 5, all 78 feature columns are in numeric form and require no categorical encoding at this stage. The absence of any additional `object` columns confirms that no typing errors were introduced by pandas during loading. It is worth noting that the two rate features `Flow Bytes/s` and `Flow Packets/s`, despite being typed as `float64`, are conceptually throughput quantities. Their floating-point typing is a consequence of the presence of `Inf` values in those columns, which are observed when `Flow Duration` equals zero. This observation is documented here and examined further in the Data Quality Verification section.

5.6 Duplicate Column

A column named `Fwd Header Length.1` is found to be an exact duplicate of `Fwd Header Length` across all 2 830 743 rows. The suffix `.1` is appended automatically by pandas when a CSV file contains two identically named columns, a known artefact of early CICFlowMeter versions. This redundant column carries no additional information and will be removed during the Data Preparation phase.

5.7 Feature Description

The two following subsections document the semantic interpretation of the 78 numeric features and address a conceptual aspect of the dataset that directly influences the modelling strategy.

5.7.1 Feature Groups

All 78 numeric features are produced by CICFlowMeter from raw packet captures. Each row represents a single bidirectional network flow, defined as the complete set of packets exchanged between two endpoints sharing the same 5-tuple (source IP, destination IP, source port, destination port, and protocol) within a session window. The features are statistical aggregates computed over all packets belonging to that flow. Table 6 presents the feature groups and their semantics.

Table 6: Feature groups in CIC-IDS2017 and their semantics.

Group	**Representative features**	**Captured information**
Flow metadata	`Flow Duration`, `Protocol`	Total session duration and transport protocol (TCP, UDP, ICMP).
Packet counts	`Total Fwd Packets`, `Total Backward Packets`	Number of packets transmitted in each direction.
Packet lengths	`Fwd Packet Length Max/Min/Mean/Std`	Statistical distribution of packet sizes in bytes.
Inter-arrival times	`Flow IAT Mean`, `Fwd IAT Total`	Time elapsed between consecutive packets; captures the temporal rhythm of traffic.
TCP flags	`SYN Flag Count`, `ACK Flag Count`, `FIN Flag Count`	Frequency of TCP control flags; characterises connection establishment and teardown.
Flow rates	`Flow Bytes/s`, `Flow Packets/s`	Throughput metrics. Subject to undefined values when `Flow Duration` equals zero.
Header lengths	`Fwd Header Length`, `Bwd Header Length`	Protocol header overhead per direction, in bytes.
Window size	`Init_Win_bytes_forward /backward`	Initial TCP receive window advertised at connection establishment.

Active / Idle times	`Active Mean`, `Idle Mean`	Duration of active transmission periods and idle intervals within the flow.
Bulk transfer	`Fwd Avg Bytes/Bulk`, `Bwd Avg Bulk Rate`	Bulk transfer statistics. All values are zero in this dataset due to a known CICFlowMeter limitation.
Target	`Label`	Class label: `BENIGN` or the name of the attack type.

As illustrated in Table 6, the feature set covers a broad spectrum of network flow characteristics. The notation **Fwd** refers to the client-to-server direction, while **Bwd** refers to the server-to-client direction.

5.7.2 Representation of Multi-Event Attacks at the Flow Level

A distinction must be drawn between the operational definition of certain attack types and the granularity at which the dataset represents them. Some attacks produce a detectable anomaly within a single network interaction: an SQL injection may manifest in the payload of a single HTTP request/response exchange. However, several attack families are inherently multi-event phenomena whose defining characteristic is a pattern that emerges across many flows, not within any single one. Three such cases are present in CIC-IDS2017:

- **DDoS**: a large volume of connection attempts directed at a single victim within a short time window. No individual flow is necessarily abnormal; the threat resides in the volume and concurrence across flows.
- **PortScan**: a systematic sequence of connection attempts across many destination ports. A single probe flow may appear entirely benign; the scan pattern only emerges when many flows are observed together.
- **Brute Force (FTP-Patator, SSH-Patator)**: repeated authentication attempts across multiple sessions. The attack is characterised by its rate and repetition across flows, not by any single authentication exchange.

Because CIC-IDS2017 is produced in a controlled experimental environment where the execution schedule of each attack is known, all flows captured during an attack window are labelled with the corresponding class. A flow-level classifier therefore learns per-flow statistical fingerprints correlated with each attack class, such as characteristic packet rates, flag distributions, or inter-arrival time patterns that emerge at the individual flow level during an attack campaign. The full operational definition of multi-event attacks, which requires aggregating evidence across many flows over time, is not directly captured at this granularity. In a production context, per-flow model scores would consequently be consumed by an alerting layer applying temporal aggregation and thresholding to produce actionable incident alerts.

6 Data Exploration

This section explores the distribution of classes and the pairwise correlations between features. All analyses are performed on the raw, uncleaned dataset; findings are therefore

preliminary and will be revisited after the cleaning steps of the Data Preparation phase.

6.1 Class Distribution

The following subsections examine the distribution of the target variable at two levels: the global distribution across the entire combined dataset, and the per-file breakdown by capture session.

6.1.1 Global Class Distribution

table 7 presents the sample count, percentage, and learnability rating for each of the 15 classes present in the combined dataset. The learnability rating is defined as follows: *Sufficient* for classes with at least 10,000 samples, *Low* for classes between 1,000 and 9,999 samples, *Insufficient* for classes between 100 and 999 samples, and *Critical* for classes with fewer than 100 samples.

Table 7: Global class distribution with learnability assessment.

Class	**Count**	**Percentage**	**Learnability**
BENIGN	2,273,097	80.300%	Sufficient*
DoS Hulk	231,073	8.163%	Sufficient
PortScan	158,930	5.614%	Sufficient
DDoS	128,027	4.523%	Sufficient
DoS GoldenEye	10,293	0.364%	Sufficient
FTP-Patator	7,938	0.280%	Low
SSH-Patator	5,897	0.208%	Low
DoS slowloris	5,796	0.205%	Low
DoS Slowhttptest	5,499	0.194%	Low
Bot	1,966	0.069%	Low
Web Attack - Brute Force	1,507	0.053%	Low
Web Attack - XSS	652	0.023%	Insufficient
Infiltration	36	0.001%	Critical
Web Attack - Sql Injection	21	0.001%	Critical
Heartbleed	11	0.000%	Critical
Total	**2,830,743**	**100%**	
* BENIGN has sufficient samples but its 80.3% dominance creates class imbalance.			

As shown in Table 7, the dataset presents a severe class imbalance: BENIGN traffic accounts for 80.3% of all rows while the 14 attack classes combined represent only 19.7%, yielding an imbalance ratio of 4.1 benign samples for every attack sample. Attack class sizes span four orders of magnitude, from 231,073 samples for DoS Hulk down to 11 for Heartbleed. Three classes fall in the Critical category, namely Infiltration, Web Attack - Sql Injection, and Heartbleed, for which standard supervised learning is unlikely to generalise reliably. La Figure 1 presents the full distribution on both linear and logarithmic scales, and la Figure 2 focuses exclusively on the attack classes to make the imbalance across minor classes visible.

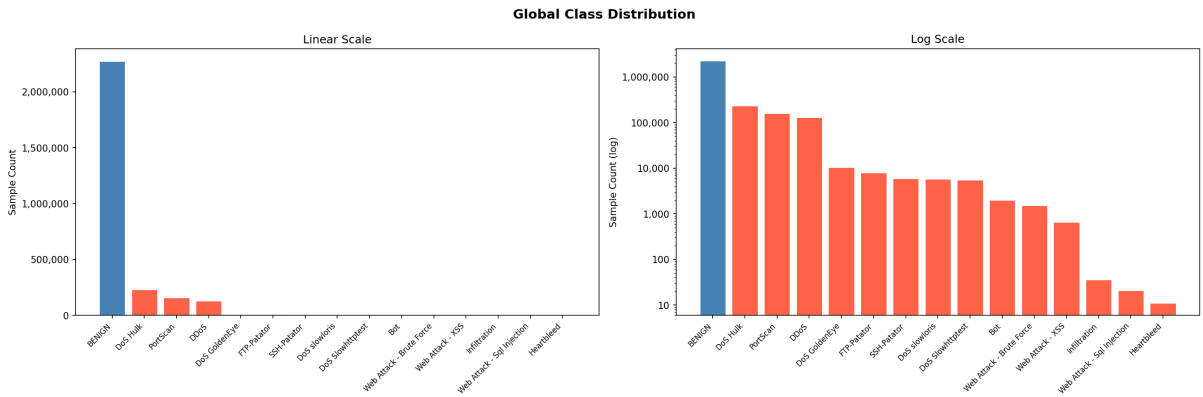

Figure 1: Global class distribution across all 15 classes (linear and log scales).

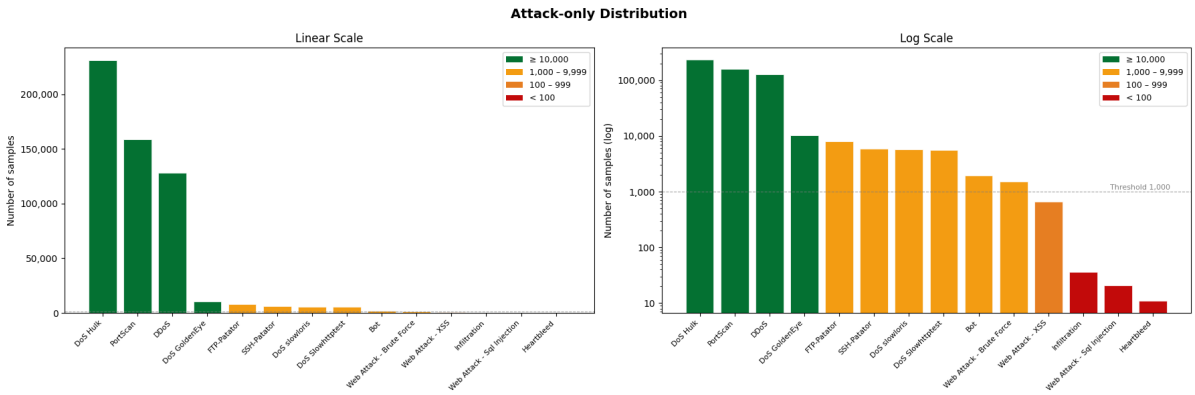

Figure 2: Attack-only class distribution with learnability colour coding.

6.1.2 Per-File Class Distribution

La Figure 3 presents the class distribution broken down by capture file, and la Figure 4 presents the BENIGN-to-attack ratio per file.

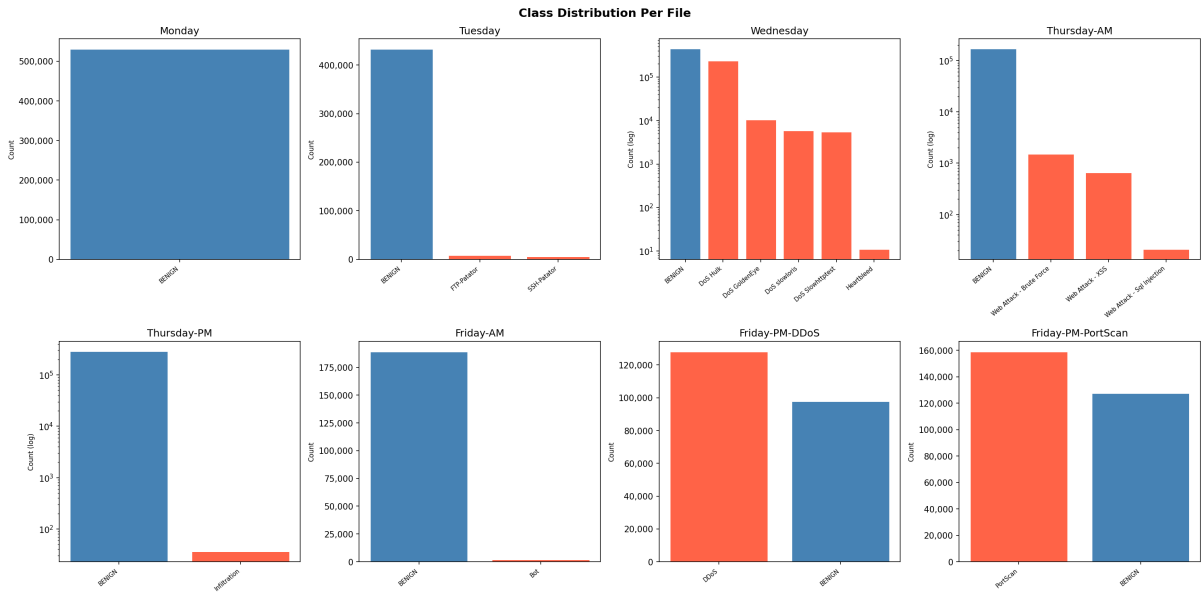

Figure 3: Class distribution per capture file.

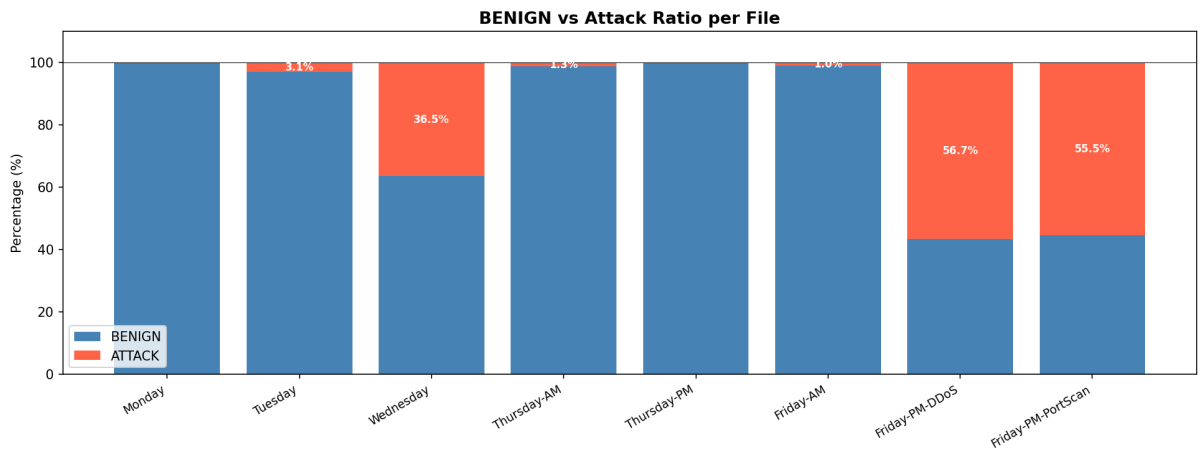

Figure 4: BENIGN vs attack ratio per file.

As illustrated in Figure 3 and Figure 4, each attack type is confined to exactly one capture file. FTP-Patator and SSH-Patator appear exclusively in Tuesday, Web Attack variants exclusively in Thursday-AM, and DDoS exclusively in Friday-PM-DDoS. This structural property has a direct consequence for evaluation strategy: a file-based train/test split would entirely exclude certain attack classes from training, making the model unable to recognise them at evaluation time. The BENIGN-to-attack ratio also varies substantially across files, from 100% benign on Monday to approximately equal proportions in certain afternoon sessions, a heterogeneity that may introduce source bias if not handled carefully during splitting. The implications for split strategy selection are examined in Section 7.

6.2 Feature Correlation

A pairwise Pearson correlation matrix is computed on the 78 numeric features of the combined dataset. Table 8 presents a selection of the 39 feature pairs whose absolute

correlation exceeds the threshold of 0.95.

Table 8: Selection of highly correlated feature pairs (absolute correlation above 0.95).

Feature A	**Feature B**	**Correlation**
Bwd Packet Length Mean	Avg Bwd Segment Size	1.000000
Fwd PSH Flags	SYN Flag Count	1.000000
Total Fwd Packets	Subflow Fwd Packets	1.000000
Fwd Packet Length Mean	Avg Fwd Segment Size	1.000000
Total Backward Packets	Subflow Bwd Packets	1.000000
Fwd Header Length	Fwd Header Length.1	1.000000
Total Length of Bwd Packets	Subflow Bwd Bytes	1.000000
Total Length of Fwd Packets	Subflow Fwd Bytes	0.999999
Flow Duration	Fwd IAT Total	0.998560
Packet Length Mean	Average Packet Size	0.997790
Flow Packets/s	Fwd Packets/s	0.987562
... 28 additional pairs above 0.95 (full list available in the notebook).		

As shown in Table 8, many of the perfect correlations reflect algebraic relationships within CICFlowMeter's feature computation: for instance, `Avg Fwd Segment Size` is by definition equal to `Fwd Packet Length Mean`, and subflow packet counts are identical to total packet counts under standard CICFlowMeter settings. These redundant features carry no additional information and are candidates for removal during the feature selection step of the Data Preparation phase. La Figure 5 presents the correlation heatmap restricted to the features involved in at least one high-correlation pair.

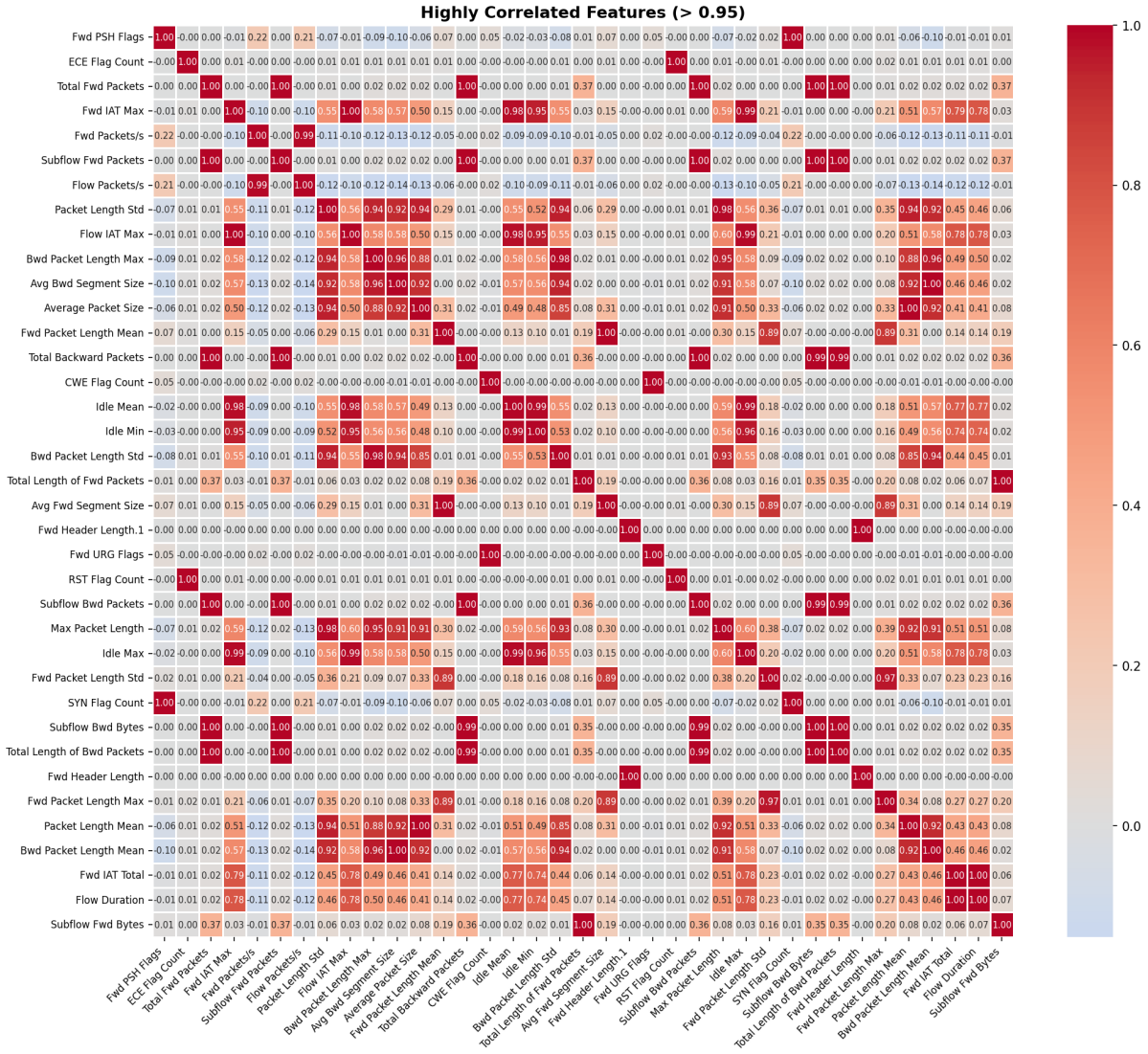

Figure 5: Pearson correlation heatmap for the subset of features involved in high-correlation pairs.

7 Data Quality Verification

This section assesses the quality of the raw combined dataset across seven dimensions: missing and infinite values, duplicate rows, constant and near-constant features, feature distribution properties, outlier quantification, feature-target association, per-file distribution drift, and finally the data leakage risk and split strategy selection. All checks are performed on the uncleaned data; the findings presented here define the cleaning agenda for the Data Preparation phase.

7.1 Missing and Infinite Values

The two following subsections document the column-level extent of missing and infinite values and assess their impact at the class level to determine whether dropping affected rows would introduce survivorship bias against rare attack classes.

7.1.1 Column-Level Analysis

Missing values (NaN) and infinite values (Inf) are examined separately. The NaN check reveals that **1,358 rows** contain at least one missing value, representing 0.0480% of the dataset. The affected column is exclusively `Flow Bytes/s`, whose missing values arise from a division by zero when `Flow Duration` equals zero: CICFlowMeter attempts to compute flow rate as bytes divided by duration, producing an undefined result for instantaneous flows. The Inf check reveals that **2,867 rows** contain at least one infinite value (0.1013% of the dataset), affecting two columns: `Flow Packets/s` (2,867 Inf entries) and `Flow Bytes/s` (1,509 Inf entries), both arising from the same root cause. Table 9 summarises these findings.

Table 9: Summary of NaN and Inf values per affected column.

Column	NaN Count	Inf Count	% of rows
`Flow Bytes/s`	1,358	1,509	0.0480% (NaN) / 0.0533% (Inf)
`Flow Packets/s`	0	2,867	0.1013%

As shown in Table 9, the combined row-level impact of NaN and Inf values is small in absolute terms, with fewer than 0.15% of all rows affected.

7.1.2 Class-Level Impact

Although the overall proportion of affected rows is small, it is necessary to verify that NaN and Inf values are not concentrated in rare attack classes, in which case dropping these rows could disproportionately reduce the already limited sample size of those classes. La Figure 6 presents the class-level impact of NaN and Inf rows.

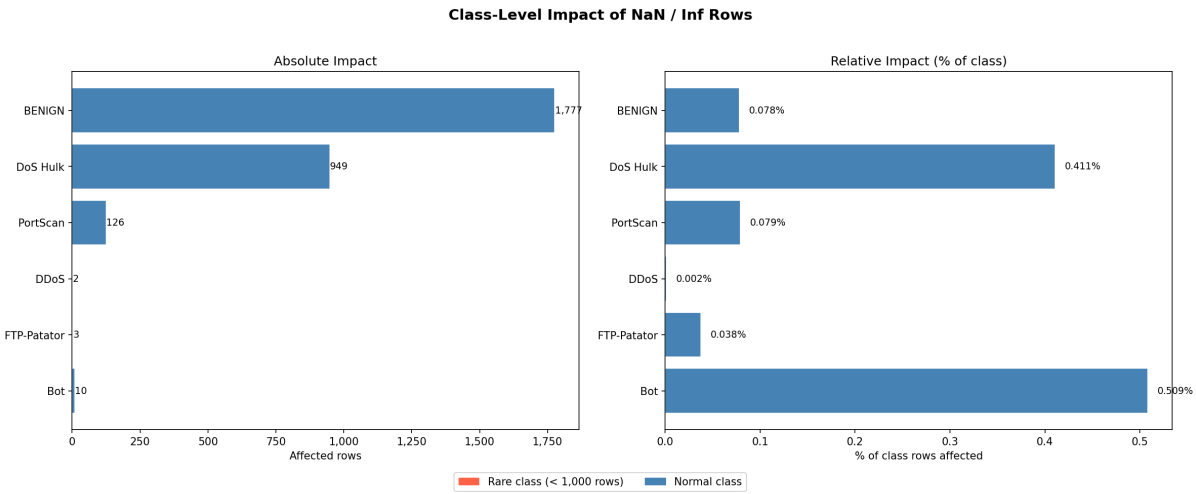

Figure 6: Class-level impact of NaN and Inf rows in absolute count and as a percentage of each class total.

As illustrated in Figure 6, the three Critical classes (Heartbleed, Web Attack - Sql Injection, Infiltration) and all Insufficient classes are entirely unaffected, with zero NaN or Inf rows. The class with the highest relative impact is Bot, which loses 10 rows out of 1,966 (0.51%), a negligible proportion. Dropping rows containing NaN or Inf values is therefore a safe strategy that introduces no survivorship bias against rare classes.

7.2 Duplicate Rows

Duplicate row analysis is conducted in two stages: an exact duplicate check, which identifies rows that are identical in all 79 columns including the label, and a contradictory duplicate check, which identifies rows whose feature vectors are identical but whose labels differ.

7.2.1 Exact Duplicates

The exact duplicate check reveals that **308,381 rows** are exact duplicates of an earlier row, representing 10.8940% of the combined dataset. Table 10 presents the distribution of duplicate rows by class.

Table 10: Exact duplicate rows by class.

Class	**Duplicate Rows**
BENIGN	176,613
PortScan	68,111
DoS Hulk	58,224
SSH-Patator	2,678
FTP-Patator	2,005
DoS slowloris	411
DoS Slowhttptest	271
Web Attack - Brute Force	37
Bot	13
DDoS	11
DoS GoldenEye	7
Total	**308,381**

As shown in Table 10, duplicate rows are common in network traffic datasets because many flows share identical statistical profiles, for instance short single-packet DNS queries. Exact duplicates in both features and label can however inflate accuracy metrics during evaluation by placing near-identical flows in both the training and test partitions. The decision on whether to deduplicate and at what granularity is deferred to the Data Preparation phase.

7.2.2 Contradictory Duplicates

A contradictory duplicate is defined as a group of rows that share an identical feature vector (verified via hash-based comparison) but are assigned different class labels. The analysis identifies **698 such groups** involving a total of **7,020 rows**. Table 11 presents the classes involved and their respective row counts within contradictory groups.

Table 11: Classes involved in contradictory duplicate groups.

Class	Rows in conflicting groups
DoS Hulk	5,224
PortScan	1,053
BENIGN	736
DDoS	6
DoS slowloris	1
Total	**7,020**

As shown in Table 11, all classes involved in contradictions are well-represented (above 1,000 samples), and no Critical or Insufficient class is affected. The origin of these contradictions lies in the flow-aggregation process: CICFlowMeter computes statistical summaries over all packets in a flow, and two flows from different sessions (one benign, one malicious) can produce statistically identical feature vectors if their traffic patterns are indistinguishable at the aggregation level. This constitutes an irreducible source of label noise. No classifier can correctly label all rows in a contradictory group from features alone, and accuracy metrics will therefore have a hard ceiling for the affected classes. All rows belonging to contradictory groups will be dropped during Data Preparation, as they carry no recoverable signal.

7.3 Constant and Near-Constant Features

The following subsections identify features that carry no discriminative information due to an absence of variation across the entire dataset, distinguishing between features that are strictly constant and those that are dominated by a single value.

7.3.1 Strictly Constant Features

Eight numeric columns take a single unique value (zero) across all 2,830,743 rows. These columns are listed in Table 12.

Table 12: Strictly constant features and their unique value.

Feature	Unique value
`Bwd PSH Flags`	0
`Bwd URG Flags`	0
`Fwd Avg Bytes/Bulk`	0
`Fwd Avg Packets/Bulk`	0
`Fwd Avg Bulk Rate`	0
`Bwd Avg Bytes/Bulk`	0
`Bwd Avg Packets/Bulk`	0
`Bwd Avg Bulk Rate`	0

As shown in Table 12, all eight constant columns are bulk transfer statistics. These features are intended to measure large continuous data transfers where the sender pushes data in bulk without waiting for acknowledgement between segments. In practice, a known bug in CICFlowMeter causes these values to be output as zero regardless of the actual traffic, a limitation documented in the research community for this specific tool

and dataset version. These features carry zero information and will be removed unconditionally during Data Preparation.

7.3.2 Near-Constant Features

Four additional features are found to be near-constant, defined as having a single dominant value in at least 99.5% of all rows. Table 13 presents these features along with their dominance percentages.

Table 13: Near-constant features with their dominant value and dominance rate.

Feature	Dominant value	Dominance (%)	Unique values
`Fwd URG Flags`	0	99.989%	2
`CWE Flag Count`	0	99.989%	2
`RST Flag Count`	0	99.976%	2
`ECE Flag Count`	0	99.976%	2

As illustrated in Table 13, the non-zero values for all four near-constant features appear exclusively in BENIGN traffic (315 rows for `Fwd URG Flags` and `CWE Flag Count`, 686 rows for `RST Flag Count`, and 689 rows for `ECE Flag Count`), meaning these features provide no signal for distinguishing any attack class from benign traffic. They are strong candidates for removal during feature selection in the Data Preparation phase.

7.4 Feature Distribution Properties

This section characterises the statistical distribution of the 78 numeric features across four dimensions: skewness, visual separation between benign and attack flows, zero inflation, and scale heterogeneity.

7.4.1 Skewness

Among the 78 numeric features, **41 exhibit a skewness with absolute value greater than 5**, indicating heavily right-tailed distributions. This is characteristic of network traffic data, where the majority of flows are short and low-volume while a small fraction involves very large or very long exchanges. La Figure 7 presents the skewness values for all 78 features, with those exceeding the threshold highlighted in red.

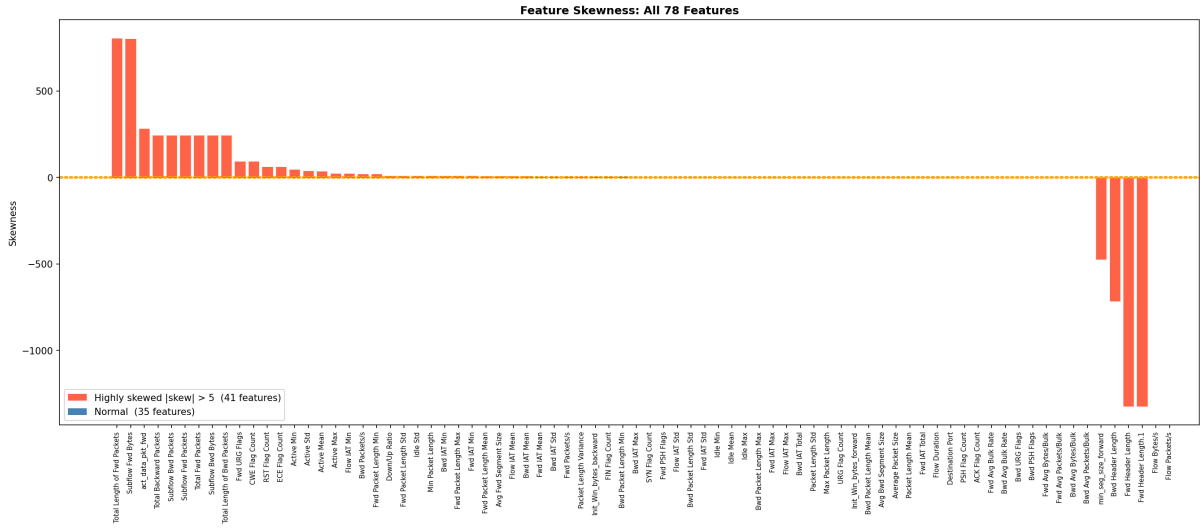

Figure 7: Skewness of all 78 numeric features; features with absolute skewness above 5 are highlighted in red (41 features).

It is noted that skewness does not invalidate tree-based models such as XGBoost, which are invariant to monotonic transformations of the input. However, highly skewed features may reduce interpretability and pose challenges for distance-based or linear models. Logarithmic or power transforms will be evaluated empirically during the Data Preparation phase.

7.4.2 Feature Distributions by Class

La Figure 8 presents overlaid histograms for eight key features, contrasting the distributions of BENIGN and ATTACK flows on the combined dataset. Values are clipped to the [q1, q99] range for visualisation clarity.

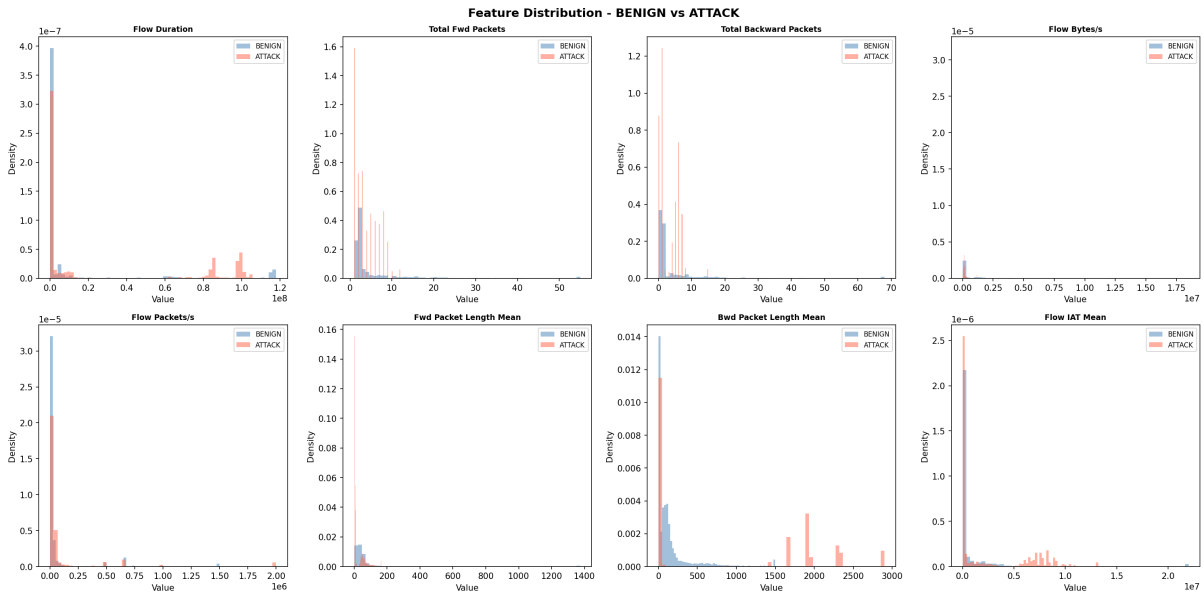

Figure 8: Distributions of eight key features for BENIGN versus ATTACK flows, clipped to the [q1, q99] range.

As illustrated in Figure 8, `Flow Bytes/s` and `Flow Packets/s` exhibit distinct peaks for the two classes, suggesting partial separability in those dimensions. Forward and backward packet length means show different profiles for attack flows, particularly for DoS-type attacks that generate many small packets. The inter-arrival time mean (`Flow IAT Mean`) also differs between classes, as attacks with rapid packet bursts produce significantly shorter inter-arrival times than benign traffic. These observations are based on binary grouping and serve as a preliminary indication of feature utility; multi-class patterns will be examined during the modelling phase.

7.4.3 Feature Scale Heterogeneity

Feature value ranges span vastly different magnitudes across the 78 numeric columns, from binary-like features taking values in $\{0, 1\}$ to flow rates and byte counts exceeding hundreds of millions. This heterogeneity is inherent to the CICFlowMeter feature set, which mixes packet counts, durations expressed in microseconds, byte sizes, and dimensionless ratios. Notably, `Fwd Header Length` and its duplicate `Fwd Header Length.1` exhibit anomalous negative minimum values ($-32{,}212{,}234{,}632$), which are likely integer overflow artefacts in CICFlowMeter and will be investigated during Data Preparation. For the planned XGBoost baseline model, feature scaling is not required as tree-based models are invariant to the scale of input features. Scaling would however be necessary if distance-based or linear models are explored in later phases.

7.5 Outlier Analysis

An outlier is defined, for each feature, as any value falling below the 1st percentile or above the 99th percentile of that feature's distribution (computed after replacing Inf values with NaN). Table 14 presents the top 10 features by total outlier count.

Table 14: Top 10 features by outlier count outside the [q1, q99] range.

Feature	**Below q1**	**Above q99**	**Total (%)**
`Fwd Packets/s`	28,308	27,618	55,926 (1.98%)
`Destination Port`	17,568	28,302	45,870 (1.62%)
`Flow IAT Mean`	3,478	28,186	31,664 (1.12%)
`Flow Duration`	2,982	28,308	31,290 (1.11%)
`Flow IAT Min`	2,891	28,308	31,199 (1.10%)
`Flow IAT Max`	2,981	28,140	31,121 (1.10%)
`Fwd Header Length.1`	1,809	26,845	28,654 (1.01%)
`Fwd Header Length`	1,809	26,845	28,654 (1.01%)
`Flow Packets/s`	28,279	334	28,613 (1.01%)
`Fwd IAT Min`	17	28,306	28,323 (1.00%)

As shown in Table 14, the class-level breakdown of outliers reveals that for `Destination Port`, FTP-Patator contributes 7,937 outlier rows out of its total 7,938 samples, indicating that extreme port values are a defining characteristic of this attack rather than noise. By contrast, for features such as `Flow IAT Mean` and `Flow Duration`, BENIGN traffic accounts for the majority of outlier rows, reflecting the natural long-tailed behaviour of normal traffic. Outlier handling during Data Preparation must therefore be informed by this class breakdown: outliers characteristic of a specific attack class should be preserved,

while those attributable to normal traffic variability may be candidates for clipping or winsorisation.

7.6 Feature-Target Association

Mutual information is computed between each of the 78 features and a binary target (BENIGN vs ATTACK) on a stratified subsample of 50,000 rows, drawn from the 2,827,876 rows that remain after removing NaN and Inf values. La Figure 9 presents the 20 features with the highest mutual information scores.

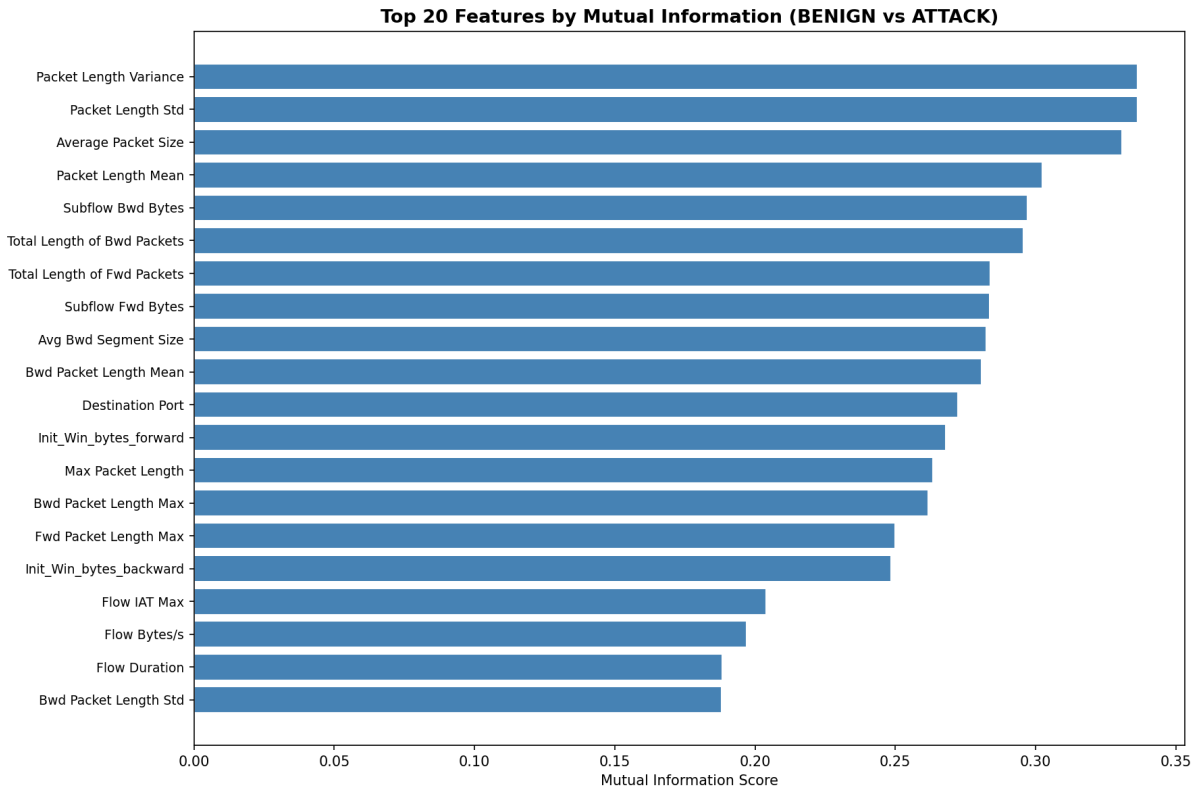

Figure 9: Top 20 features by mutual information score with respect to the binary BENIGN vs ATTACK target.

As illustrated in Figure 9, packet length statistics emerge as the most discriminative features: `Packet Length Variance` and `Packet Length Std` both score 0.3364, followed by `Average Packet Size` at 0.3307. These results are consistent with domain knowledge, as attacks such as DoS and DDoS alter the packet size distribution measurably at the flow level. At the opposite extreme, all bulk transfer features score at or near zero (below 0.0002), consistent with the finding that these features are always zero due to the CICFlowMeter limitation documented in Section 7.3.1. It is noted that these scores reflect association with a binary target only; feature importance rankings may differ for the multi-class classification task.

7.7 Per-File Feature Distribution Drift

To isolate the effect of temporal and environmental variation from the presence of attacks, the median values of six representative features are compared across the eight capture

files using BENIGN-only traffic. Any observed shift is attributable exclusively to session-specific factors such as time of day, network load, or traffic composition.

The analysis reveals substantial drift for certain features: `Flow Duration` exhibits a maximum-to-minimum median ratio of **560.09** across files, flagged as high drift, driven by the distinctly shorter flows observed in the Thursday-PM session (median of 306 microseconds vs. over 171,000 in Friday-PM-DDoS). By contrast, `Total Fwd Packets` remains stable across all files with a median of 2 in every session, indicating that this feature is robust to session-level variation. Features with high drift ratios may cause the model to learn session-specific patterns rather than genuine attack signatures, and will be monitored during cross-validation in the modelling phase.

7.8 Data Leakage Risk and Split Strategy

The following two aspects of data leakage are examined before selecting a split strategy. First, the presence of non-target object-type features is checked: no such columns exist beyond the `Label` column itself, confirming that no string-valued identifier or metadata column can leak class information into the feature space. Second, contradictory duplicate groups are confirmed to be present (698 groups, as documented in Section 7.2.2), introducing an irreducible Bayes error for any classifier.

Table 15 presents the trade-offs between five candidate splitting strategies for CIC-IDS2017.

Table 15: Comparison of candidate train/test split strategies for CIC-IDS2017.

Strategy	**All classes?**	**Leakage risk**	**Complexity**
Random	Not guaranteed	High	Low
File-based (by day)	No	Low	Low
Stratified split	Yes	Medium	Easy
Stratified K-Fold	Yes	Medium	Medium
Temporal + stratified/file	Yes	Low	Hard

As shown in Table 15, two strategies are immediately excluded by structural constraints. The random split does not guarantee that all 14 attack types appear in both training and test partitions, which is particularly problematic for Critical classes such as Heartbleed (11 samples), and risks placing temporally adjacent flows from the same session in both partitions, inflating performance metrics. The file-based split fails because each attack type is confined to exactly one file: any file excluded from training renders an entire attack class unseen by the model. Stratified K-Fold is therefore selected as the recommended primary evaluation strategy, as it guarantees proportional representation of every class in each fold while providing multiple evaluation runs for more robust metric estimation. Implementation is deferred to the Data Preparation phase.

8 Conclusion

This report presented the Data Understanding phase of the CRISP-DM methodology applied to the CIC-IDS2017 dataset. The analysis was conducted on the raw, uncleaned data and covered four successive tasks: data loading, data description, data exploration, and data quality verification.

The dataset was found to comprise approximately 2.83 million flow-level records across 79 columns, representing 15 traffic classes generated over five working days. The examination of the class distribution revealed a severe imbalance, with BENIGN traffic accounting for 80.3% of all samples, while three attack classes (Heartbleed, Infiltration, and Web Attack - Sql Injection) each contained fewer than 100 samples, placing them in the Critical learnability category. The most consequential structural property identified was that each attack type is confined to exactly one capture file, a constraint that eliminated file-based splitting as a viable evaluation strategy and motivated the adoption of Stratified K-Fold cross-validation.

The data quality verification identified several issues that must be addressed before modelling. NaN and Inf values were found to be concentrated in two rate features (`Flow Bytes/s` and `Flow Packets/s`), affecting fewer than 0.15% of rows with no bias against rare classes. Exact duplicate rows represented 10.89% of the dataset, and 698 contradictory duplicate groups were identified, introducing an irreducible source of label noise. Eight features were found to be strictly constant across all rows due to a known CICFlowMeter bug, and four additional features were near-constant, both groups being candidates for unconditional removal. Feature analysis further revealed pervasive right-skewness in 41 out of 78 features, 39 highly correlated feature pairs reflecting algebraic redundancies in the CICFlowMeter feature set, and substantial per-file distribution drift in BENIGN traffic confirming the presence of temporal variation across capture sessions.

It is noted that all statistics computed in this phase, including correlation coefficients, skewness values, and descriptive summaries, were computed on raw data containing Inf values and are therefore preliminary. All findings will be recomputed on cleaned data following the Data Preparation phase.

The Data Preparation phase will address the identified issues in the following order: replacement of Inf values with NaN followed by row removal, elimination of contradictory and exact duplicate rows, removal of constant and near-constant features, resolution of highly correlated feature pairs through feature selection, label encoding, implementation of Stratified K-Fold splitting, treatment of class imbalance, and normalisation of numerical features where required by the chosen model family.